

8.3 Classwork: Rational Functions as Transformations — Markscheme

Parent function $f(x) = \frac{1}{x}$: vertical asymptote $x = 0$, horizontal asymptote $y = 0$.

1. Translations.

(a) $g(x) = \frac{1}{x+4}$: **horizontal translation 4 units left** (translation by $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$).

(b) $h(x) = \frac{1}{x} + 3$: **vertical translation 3 units up** (translation by $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$).

(c) $p(x) = \frac{1}{x-2} + 5$: translation **2 units right and 5 units up** (by $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$). **VA:** $x = 2$;
HA: $y = 5$.

(d) y -intercept: $p(0) = \frac{1}{-2} + 5 = \frac{9}{2} = 4.5$, point $(0, \frac{9}{2})$.

Right branch ($x > 2$): from $+\infty$ at $x = 2^+$ down to $y = 5$ from above; through $(3, 6)$, $(4, 5.5)$, $(7, 5.2)$.

Left branch ($x < 2$): from $y = 5$ from below at $x \rightarrow -\infty$ down to $-\infty$ at $x = 2^-$; through $(1, 4)$, $(0, \frac{9}{2})$, $(-2, 4.75)$.

2. Reflections and stretches.

(a) Reflection in the x -axis: $y = -\frac{1}{x}$.

(b) Vertical stretch, scale factor 2: $y = \frac{2}{x}$.

(c) $g(x) = \frac{-2}{x-1}$.

(i) **Vertical stretch factor 2 and reflection in the x -axis** (together giving $y = -\frac{2}{x}$), then **translation 1 unit right**. *Accept any order in which the stretch/reflection precede the horizontal translation.*

(ii) **VA:** $x = 1$; **HA:** $y = 0$.

(iii) x -intercept: $-\frac{2}{x-1} = 0$ has no solution $\Rightarrow$ **no x -intercept**. y -intercept:
 $g(0) = \frac{-2}{-1} = 2$, point $(0, 2)$.

(iv) Branch $x > 1$: below the x -axis, from $-\infty$ at $x = 1^+$ rising to 0^- ; through $(2, -2)$, $(3, -1)$, $(5, -0.5)$.

Branch $x < 1$: above the x -axis, from 0^+ at $x \rightarrow -\infty$ rising to $+\infty$ at $x = 1^-$; through $(0, 2)$, $(-1, 1)$, $(-3, 0.5)$.

3. **From $\frac{2x+7}{x+3}$ to a transformation.**

(a) $2x+7 = 2(x+3) + 1$, so

$$f(x) = \frac{2(x+3) + 1}{x+3} = 2 + \frac{1}{x+3}.$$

(b) Transformations of $y = \frac{1}{x}$: **translation 3 units left and 2 units up** (by $(\begin{smallmatrix} -3 \\ 2 \end{smallmatrix})$); no stretch, since the coefficient of $\frac{1}{x+3}$ is 1.

(c) **VA:** $x = -3$; **HA:** $y = 2$. x -intercept: $2x+7 = 0 \Rightarrow x = -\frac{7}{2}$, point $(-\frac{7}{2}, 0)$.
 y -intercept: $f(0) = \frac{7}{3}$, point $(0, \frac{7}{3}) \approx (0, 2.33)$.

(d) Right branch ($x > -3$): from $+\infty$ at $x = -3^+$ down to $y = 2$ from above; through $(-2, 3)$, $(0, \frac{7}{3})$.

Left branch ($x < -3$): from $y = 2$ from below at $x \rightarrow -\infty$ down to $-\infty$ at $x = -3^-$; through $(-\frac{7}{2}, 0)$, $(-4, 1)$, $(-5, 1.5)$.

4. **Building a function from transformations.**

(a) Stretch $\times 3$: $\frac{3}{x}$; then right 2, down 1:

$$g(x) = \frac{3}{x-2} - 1.$$

(b) **VA:** $x = 2$; **HA:** $y = -1$.

(c) y -intercept: $g(0) = \frac{3}{-2} - 1 = -\frac{5}{2}$, point $(0, -\frac{5}{2})$.

x -intercept: $\frac{3}{x-2} - 1 = 0 \Rightarrow \frac{3}{x-2} = 1 \Rightarrow x-2 = 3 \Rightarrow x = 5$, point $(5, 0)$.

(d) $g(8) = \frac{3}{8-2} - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$.

5. **Quick identification.**

(a) $y = \frac{1}{x-5}$: translation 5 units right. **VA** $x = 5$, **HA** $y = 0$.

(b) $y = \frac{1}{x} - 4$: translation 4 units down. **VA** $x = 0$, **HA** $y = -4$.

(c) $y = \frac{4}{x}$: vertical stretch, scale factor 4 (*accept: horizontal stretch, scale factor 4*).
VA $x = 0$, **HA** $y = 0$.

(d) $y = -\frac{1}{x}$: reflection in the x -axis (*accept: reflection in the y -axis — both give the same graph*). **VA** $x = 0$, **HA** $y = 0$.

(e) $y = \frac{1}{x+2} + 1$: translation 2 units left and 1 unit up. **VA** $x = -2$, **HA** $y = 1$.

6. Transforming a given graph.

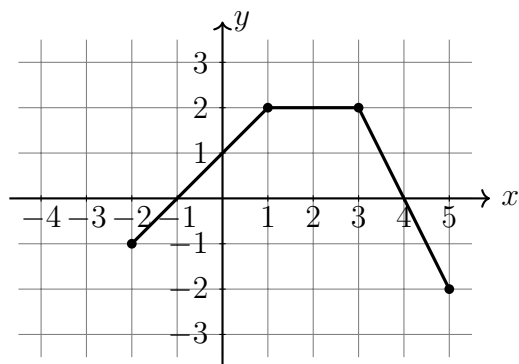
(a) $f(0) = 2$ and $f(2) = 2$.

(b) $g(x) = f(x) - 3$: **vertical translation 3 units down**. Image of $(0, 2)$ is $(0, -1)$.

(c) $y = f(x - 1)$: **translation 1 unit right**. Images of the labelled points:

$$(-3, -1) \rightarrow (-2, -1), \quad (0, 2) \rightarrow (1, 2), \quad (2, 2) \rightarrow (3, 2), \quad (4, -2) \rightarrow (5, -2),$$

joined by straight segments in the same order.



(Solution graph for 6(c): $y = f(x - 1)$.)